

Surveillance tools are all means of technology provided and used by the surveillance industry, police or military intelligence, and national security institutions that enable individual surveillance and mass surveillance.

The following components are used for surveillance, some of which were originally cited by Steven Ashley in Scientific American:

Primarily electronic

Digital still and video cameras (CCTVs)

GPSs for tracking

Electronic toll collection

Computer and network surveillance

Phone tapping

Cell phone monitoring

Voice recognition, facial recognition, walking gait analysis and other biometric characteristics

Covert listening devices or "bugs", tiny, hidden microphone and short-range radio transmitter

Directional microphones

Smart glasses, and other wearable technologies

Primarily chemical

Artificial noses

Chemical markers like UV markers

DNA sensors: Biochip etc., for screening tiniest traces of body material

Other

Airplanes, unmanned aerial vehicles and satellites

Night-vision goggles or telescopes

Laser beam bounced off a window to record vibrations in the pane from conversations in the room

Discarded items containing personal information, like  
phone bills,  
credit-card statements and  
computer hard drives (using digital forensics)

The electronic means, especially when combined with Internet features (ubiquitous computing, IoT) and enhanced by artificial intelligence analysis methods readily lend themselves to mass surveillance. This is why countersurveillance measures like anonymization and end-to-end encryption have become critical. Devices like chemical markers, on the other hand are more suited and in fact designed mainly for monitoring individuals.

See also

Global surveillance disclosures (2013–present)

List of government mass surveillance projects

National Applications Office

Surveillance Detection Unit

## References

### External links

"Mass Surveillance Part 1-Risks and opportunities raised by the current generation of network services and applications". EPRS European Parliamentary Research Service Scientific Foresight (STOA) UnitPE. 2014. pdf

Ashley, Steven (September 2008). "Tools of the Spy Trade". *Scientific American*. 299 (3): 70–71.

Bibcode:2008SciAm.299c..70A. doi:10.1038/scientificamerican0908-70.